

Potential Adverse Effects of Phytoestrogens^{1,2}

PATRICIA L. WHITTEN,^{3*} CAROLE LEWIS,[†] ELIZABETH RUSSELL*
AND FREDERICK NAFTOLIN[†]

*Departments of Anthropology and Biology, Emory University, Atlanta, GA 30322; and [†]Department of Obstetrics and Gynecology, Yale University School of Medicine, New Haven, CT 02138

ABSTRACT Evaluation of the potential benefits and risks offered by naturally occurring plant estrogens requires investigation of their potency and sites of action when consumed at natural dietary concentrations. Our investigations have examined the effects of a range of natural dietary concentrations of the most potent plant isoflavonoid, coumestrol, using a rat model and a variety of estrogen-dependent tissues and endpoints. Treatments of immature females demonstrated agonistic action in the reproductive tract, brain, and pituitary at natural dietary concentrations. Experiments designed to test for estrogen antagonism demonstrated that coumestrol did not conform to the picture of a classic antiestrogen. However, coumestrol did suppress estrous cycles in adult females. Developmental actions were examined by neonatal exposure of pups through milk of rat dams fed a coumestrol, control, or commercial soy-based diet during the critical period of the first 10 postnatal days or throughout the 21 days of lactation. The 10-day treatment did not significantly alter adult estrous cyclicity, but the 21-day treatment produced in a persistent estrus state in coumestrol-treated females by 132 days of age. In contrast, the 10-day coumestrol treatments produced significant deficits in the sexual behavior of male offspring. These findings illustrate the broad range of actions of these natural estrogens and the variability in potency across endpoints. This variability argues for the importance of fully characterizing each phytoestrogen in terms of its sites of action, balance of agonistic and antagonistic properties, natural potency, and short-term and long-term effects. *J. Nutr.* 125: 771S-776S, 1995.

INDEXING KEY WORDS:

- isoflavonoids • coumestrol • phytoestrogens
- sexual differentiation • rat

The isoflavonoids are a group of plant chemicals common in legumes that are known for their estrogenic actions. Because of the broad spectrum of actions of endogenous estrogens, there is growing interest in the potential use of these chemicals in nutritional and pharmaceutical applications (Adlercreutz et al. 1982; Adlercreutz 1990; Axelsson et al. 1982, 1984). Like

steroidal estrogens, however, these plant estrogens have the potential to exert adverse as well as beneficial actions. These potential roles fall into three major areas: 1) estrogen agonists whose activational actions could prove beneficial to postmenopausal women but might be harmful to the degree that they contribute to carcinogenesis or other adverse effects; 2) antiestrogens or antiproliferative agents that could help to prevent estrogen dependent carcinoma by antagonizing estrogen action but could also contribute to infertility by suppressing normal reproductive function; and 3) developmental toxins that could disrupt sexual differentiation by altering sex-specific patterns of development but might also provide protection against environmental estrogens by altering steroid response thresholds.

These divergent outcomes make it important to fully investigate the range of their potential actions. Although the isoflavonoids and other estrogenic plant chemicals are generally lumped together under the general rubric of phytoestrogen, emerging evidence suggests that this is a diverse group with a variety of biochemical and biological properties (Whitten and Naftolin 1991). Therefore it is important to characterize the action of each phytoestrogen individually. Of particular concern, due to the interest in the role of soy in the prevention and treatment of cancer and heart disease, is the action and efficacy of isoflavonoids when consumed at normal dietary concentrations. To

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³ To whom correspondence should be addressed: Anthropology Department, 1557 Pierce Drive, Emory University, Atlanta, GA 30322.

address these issues, we have used an *in vivo*, rat model of estrogen action and a paradigm of dietary treatment to test the effects of natural dietary concentrations of isoflavonoids on a variety of tissues and processes.

MATERIALS AND METHODS

Chemicals

Sephadex LH-20 was obtained from Pharmacia Fine Chemicals (Piscataway, NJ). $17\beta[2,4,6,7-^3\text{H}]$ -estradiol ($[^3\text{H}]\text{E}_2$, 112 Ci/mmol), $[^3\text{H}]\text{R5020}$ (86.9 Ci/mmol) and unlabeled R5020 were purchased from New England Nuclear (Boston, MA). Unlabeled estradiol (E_2) and diethylstilbestrol (DES) were obtained from Sigma (St. Louis, MO). Coumestrol was obtained from Eastman Kodak Company (Rochester, NY) and Spectrum Chemical Corp. (Gardena, CA). Purity was assessed by melting point determinations.

Animals

Animal subjects were female rats, Sprague Dawley strain (Camm, Wayne, NJ, Harlan, Indianapolis, IN) and their male and female offspring. The animals were maintained on a 12-h light/dark cycle (lights on: 0630) with free access to food and water. Animal protocols were reviewed and approved by the Institutional Animal Care and Use Committees of Yale and Emory Universities.

Dietary administration of flavonoids

Because commercial soy-based rodent diets contain some isoflavonoids (Axelson et al. 1984), all animals were fed a semipurified formulation, the American Institute of Nutrition (AIN) semipurified rat-mouse diet (US Biochemical Corp., Cincinnati, OH) containing the following ingredients: casein-high nitrogen 20%, DL-methionine 0.3%, cornstarch 15%, sucrose 50%, fiber-celufil 5%, corn oil 5%, mineral mixture 3.5%, vitamin mixture 1%, choline bitartrate 0.2%. Prior experiments have shown that the phytoestrogen excretion of rats is rapidly and markedly reduced with this diet. Coumestrol and other flavonoids were incorporated into the AIN diet at a range of natural dietary concentrations, as reported below. Control animals received the AIN diet alone. Additional experiments examined the action of a commercial soy-based rodent diet (Purina Rodent Chow, Ralston, MO).

Tests of estrogen agonism and antagonism

Estrogen agonism and antagonism was assessed in immature females (20–21 days old 30–40 g). In tests of estrogen agonism, females were treated for 90 h

with experimental diets. In tests of estrogen antagonism, coumestrol was administered for 90 h before the initiation of E_2 treatment; all animals were then placed on the control diet and E_2 (0.1 $\mu\text{g}/\text{d}$ sc in 0.1 ml saline) was administered for three mornings. Uterine weights were determined on the fourth day, 24 h after the last E_2 injection.

Animals were killed by decapitation, and the entire pituitary, uterus, and brain were rapidly removed. The hypothalamic-preoptic (HPOA) was dissected as described in Whitten and Naftolin (1992). All tissues were placed in a solution of cold 5% dimethylsulfoxide (DMSO) and frozen on dry ice. Tissues were stored at -80°C until the time of assay. All subsequent procedures were conducted at $0-4^\circ\text{C}$. After weighing, uteri were minced finely and homogenized in TEGM buffer (10-mM Tris-HCL, 1.5-mM EDTA, 10% glycerol, 12-mM monothioglycerol, pH 7.4) using a ground-glass homogenizer at a dilution of 40 mg tissue/ml buffer. Brain tissue and pooled whole pituitaries (3 per observation) were homogenized in 0.6-ml TEGM using glass tissue-grinders with a Teflon pestle. Homogenates were processed for cytosolic progesterin and estrogen receptors and nuclear estrogen receptors as described in Whitten and Naftolin (1992) and Whitten et al. (1992).

Developmental studies

Developmental studies utilized timed pregnant females obtained on the 15th day of gestation. Offspring were pooled at birth and randomly assigned to dams to constitute dietary treatment groups. The lactating dams were treated for 21 (Study I) or 10 (Study II) days with an AIN control diet, commercial soy-based diet, or 100 $\mu\text{g}/\text{g}$ coumestrol diet.

Vaginal cyclicity

Following 21 day postnatal treatment, vaginal smears were taken daily by lavage on postnatal days 99–101, 105–109, 132–136, and 139–143 days. Following 10 d postnatal treatment, vaginal smears were taken on days 44–70, 128–137, 197–207, and 268–272. Animals were classified by vaginal smear pattern as follows: 1) cornified: cornified cells predominant in $>70\%$ of smears, 2) leukocytic: leukocytes predominant in $>70\%$ of smears; and 3) cyclic: no more than 70% of smears cornified or leukocytic.

LH release

At 143 days of age, rats were bilaterally ovariectomized under ether anesthesia. Three weeks later, the animals were injected subcutaneously at 0900 h with 10 μg estradiol benzoate (EB) in sesame oil (0.05 ml). Three days later, at 0900 h, blood samples were ob-

tained by cardiac puncture under ether anesthesia. At 1200 h, the animals were injected subcutaneously with 1 mg progesterone in sesame oil (0.1 ml). At 1600 h, animals were killed, and trunk blood was collected. Serum LH was measured by radioimmunoassay using reagents from the National Institute of Diabetes and Digestive and Kidney Diseases (NIDDK) rat pituitary hormone program. Antiserum to rat LH (NIDDK-anti-rLH-S-10), a ^{125}I -rat LH tracer (NIDDK rLH-I-8: AFP-11536B) and a rat LH preparation as standard (NIDDK-RP-3) were provided.

Sexual behavior

Sexual behavior was tested in intake adult males. Four to five males per treatment group were tested. Tests for male sexual behavior were performed in a plexiglass arena (12 in. diam) during the dark phase of the cycle under dim red light. Testing began 2 h after the onset of darkness. Stimulus animals were ovariectomized females unfamiliar to the male who were rendered sexually receptive by subcutaneous injection of estradiol benzoate 48 h before testing and progesterone in 0.1 ml sesame oil, 4–6 h before testing. Males were allowed to adapt the arena for 10 min before addition of the stimulus female and the start of the test. All mounts, intromissions, and ejaculations were recorded during a 30-min test period. After each test, the arena was thoroughly cleaned.

Statistical analyses

Student *t* tests were used to test the significance of differences between control and coumestrol treated animals. The χ^2 test was used to test the significance of differences in cyclicity between treatment groups. Repeated measures analysis of variance (MANOVA; SPSS/PC + 4.0) was used to test the significance of differences in LH surges between treatment groups. A significance level of $P < 0.05$ was used as the measure of significance.

RESULTS

Estrogenic and antiestrogenic actions

The balance of agonistic and antagonistic effects produced by naturally dietary exposure to phytoestrogens was examined by incorporating isoflavonoids and other flavonoids in a semipurified diet at concentrations found naturally in human and animal foods and feeding those diets to immature female rats. Several classic measures of estrogenicity were examined: uterine growth, progesterin receptor induction, nuclear estrogen receptor activation, and vaginal canalization. Uterine weight responses to flavonoid diets were in

accord with their relative estrogen receptor binding affinity (RBA) (Whitten and Naftolin 1991, Whitten et al. 1992). **Table 1** shows that a low-dose (100 $\mu\text{g/g}$) coumestrol diet was estrogenic over a variety of tissues and endpoints.

Estrogen antagonism was examined by feeding coumestrol diets during or prior to treatment with a maximal physiologic dose of estradiol (0.1 $\mu\text{g/d}$). Coumestrol did not inhibit the uterotrophic action of estrogen but did augment estradiol's down-regulatory effects on cytosolic and nuclear estrogen receptor binding (**Table 2**; Whitten et al. 1994).

Influences on estrous cyclicity

Investigations involving more prolonged treatment indicated that the balance between agonism and suppression of estrogen action was dependent on the hormonal milieu. For example, coumestrol stimulated the onset of estrous cycles when provided prepubertally to immature females (**Table 1**), but the same low-dose coumestrol diet inhibited ovarian cycles when provided to cycling adult females (**Fig. 1**).

Defeminization of gonadotropin function

Because sex steroids regulate the sexual differentiation of the central nervous system, exposure to exogenous androgens or estrogens early in development can disrupt the normal sex-specific patterns of development, resulting in females in defeminization of the preovulatory LH surge mechanisms (Whitten and Naftolin 1994). In female rats, perinatal exposure to

TABLE 1
Estrogenic effects of a low-dose coumestrol diet¹

Index	Tissue	Mean increase ²
Cell proliferation	Uterus	228%*
Progesterin receptor induction	Uterus	311%*
	Pituitary	459%*
	Hypothalamus	134%*
Estrogen receptor activation	Uterus	297%*
Age of canalization	Vagina	90%*
Age of first estrus	Vagina	89%*

¹ The concentration of coumestrol in the low dose diet was 100 $\mu\text{g/g}$ of AIN semipurified rat-mouse diet. Responses were determined following 4 days of treatment for all indices except vaginal canalization, where treatments continued from day 21 until day 60 ($N = 6-12/\text{group}$).

² Proportionate elevation over mean control values originally expressed in grams (uterine growth), fmol/mg (steroid receptors), or days (vaginal canalization and first estrus), calculated from data means published in Whitten and Naftolin, 1992; Whitten et al., 1992.

* $P < 0.05$.

TABLE 2

Effect of pretreatment with a coumestrol diet on the uterotrophic action of estradiol¹

Index	% of E ₂ response ²
Wet weight	146%*
Protein content	143%*
Cytosolic progesterin receptors	86% ns ³
Cytosolic estrogen receptors	65%*
Estrogen receptor activation	78%*

¹ A 500 µg/g coumestrol diet was provided for 4 days prior to E₂ treatment (0.1 µg/d sc for 3 days) and uterine weight and cytosolic receptor determinations were made 24 h later (N = 7/group). Estrogen receptor activation was determined 3 h after injection of 0.1 µg E₂.

² Response induced by E₂ following coumestrol pretreatment expressed as percentage of response induced by E₂ alone, calculated from data means published in Whitten and Naftolin, 1994.

³ ns = not significant.

* P < 0.05.

exogenous estrogens results in a premature anovulatory syndrome characterized by persistent cornification of the vaginal epithelia and an inability to exhibit an LH surge. Nonsteroidal estrogens are especially potent in this regard since they do not bind with high affinity to alphafetoprotein. The effect of phytoestrogens on the sexual differentiation of gonadotropin function was examined by neonatal exposure of pups through milk of rat dams fed a coumestrol (100 µg/g), control, or a commercial soy-based diet during the critical period of the first 10 postnatal days or throughout the 21 days of lactation (Whitten et al. 1993, Whitten et al. 1995). **Table 3** shows that the 10-d treatment did not significantly alter adult estrous cyclicity, but the 21-d treatment produced in a persistent estrous state in coumestrol-treated females by 132 d of age. Persistent estrous was accompanied by a failure to exhibit an LH surge at the expected time in response to estradiol plus progesterone priming, suggesting a central nervous system locus for these effects (Whitten et al. 1993).

Demasculinization of male sexual behavior

In male rats, perinatal exposure to exogenous estrogen prevents the masculinization of sexual behavior and defeminization of gonadotropin function, presumably because the estrogens either compete with locally synthesized estradiol or inhibit testicular androgen production (Whitten and Naftolin 1994). To test for phytoestrogen influences on male sexual development, male rats lactationally exposed to control, coumestrol, or commercial soy-based diets during the first 10 postnatal days were tested for the expression of male sexual behavior in adulthood. Intact males were placed in an arena with a receptive control fe-

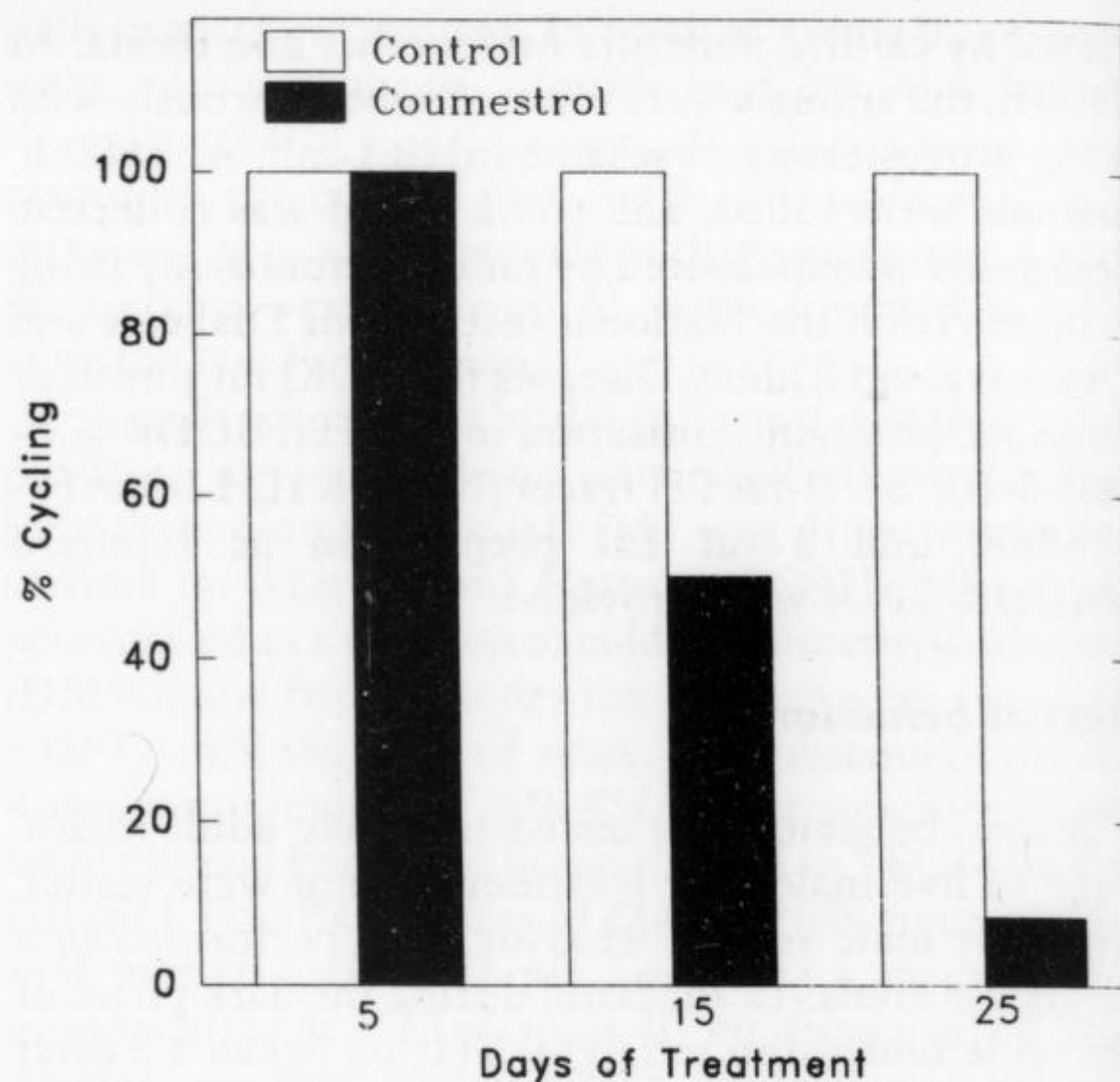


FIGURE 1 Percentage of cycling female rats (N = 12/group) following 10–25 d treatment with a control or low-dose (100 µg/g) coumestrol diet.

male, and mount and ejaculation frequencies and latencies were recorded over a 30-min period. Table 3 shows that coumestrol increased the latency to mount and ejaculate and reduced mount and ejaculation rates. Testicular weights and plasma testosterone levels did

TABLE 3

Developmental effects of lactational exposure to a low-dose coumestrol diet¹

Index	Treatment duration	% of Control ²
Female Gonadotropin Function		
Onset of persistent estrus	D1-10	100% ns ³
	D1-21	49%*
E2 + P induced LH surge	D1-21	13%*
Male Sexual Behavior		
Mount latency	D1-10	325%*
Ejaculation latency	D1-10	777%*
Mount rate	D1-10	30%*
Ejaculation rate	D1-10	24%*
Testicular size	D1-10	93% ns
Serum Testosterone	D1-10	76% ns

¹ A 100 µg/g coumestrol diet was provided to lactating rat dams (N = 2–4/group) for the first 10 or 21 postnatal days. Female offspring (N = 4–12/group) were tested for vaginal cyclicity and for LH response following ovariectomy and E + P priming. Male offspring (N = 4–5/group) were tested for sexual behavior.

² Response of coumestrol-treated offspring expressed as a percentage of control response. Calculated from data means published in Whitten et al. 1993.

³ ns = not significant.

* P < 0.05.

not differ among treatment groups, suggesting that the deficits in male sexual behavior were not due to deficits in adult gonadal function.

DISCUSSION

Dietary coumestrol meets many of the criteria of a classical estrogen but appears to fulfill few of the criteria of an antiestrogen, raising questions about the purported antiproliferative actions of isoflavonoids. Although these data do not rule out the presence of antiestrogenic actions in other isoflavonoids, they do indicate that estrogen antagonism cannot be considered to be an intrinsic phytoestrogen trait and should be established for each chemical individually. It also may be that the antiproliferative actions of isoflavonoids are not a function of estrogen antagonism since *in vitro* studies suggest that estrogen receptors are not required for the antiproliferative actions of genistein (Peterson and Barnes 1991).

The developmental actions of coumestrol also urge caution in the adoption of these chemicals for preventative and therapeutic applications. The induction of progesterin receptors in pituitary and hypothalamus and the altered sexual behavior and gonadotropin function in lactationally exposed males and females provide evidence for coumestrol action in the central nervous system. These actions indicate that a low-dose coumestrol diet can alter neuroendocrine development and in some circumstances might act as a developmental toxicant.

Further investigations will be required to determine the broader implications of these results for the use of isoflavonoid-rich products. The widespread use of soy in healthy, reproducing Asian populations would appear to provide evidence that human populations are not adversely affected by developmental exposure to these phytoestrogens. Indeed, our investigations produced no evidence that a commercial soy-based diet induced developmental disorders either, even though our previous experiments indicated that at least some batches are uterotrophic (Whitten et al. 1992). However, the available evidence suggests that isoflavonoid type and concentration, along with the duration of exposure, endpoint and subjects are important variables in determining developmental outcomes. For example, our feeding experiments focussed on coumestrol because it is the most potent estrogen of the isoflavonoids and thus might be expected to most easily provide evidence of the most extreme developmental outcomes. Coumestrol, however, appears to be present in lower concentrations than other isoflavonoids in soy (reviewed in Whitten and Naftolin 1991) and so far has not been detected in the urine of humans consuming soy. Whether that reflects its absence, its presence below the limits of detectability,

its metabolism to other forms, or differences in its extractability is as yet unresolved. However, developmental actions do not appear to be limited to coumestrol since other investigations using a paradigm of parenteral administration have found that perinatal genistein as well as coumestrol alters gonadotropin function female rats (Faber and Hughes 1991). In the latter experiments, however, the direction and significance of the influence on basal LH, GnRH-induced LH release, and volume of the sexually dimorphic nucleus varied with both isoflavonoid, dose and the endpoint examined. In our investigations, alterations of male sexual development were apparent following a shorter-term treatment than that required to influence female sexual development, which could reflect either sex differences in developmental responsiveness to phytoestrogens or differences in the lability of sexual behavior vs. gonadotropin function. Because at least some of these outcomes reflect variability in thresholds of response to estrogen, it will be important to determine the import of this variability in endpoint response for adverse vs. potentially beneficial developmental outcomes, such as potential resistance to the carcinogenic action of environmental estrogens.

The emerging picture of variability in isoflavonoid action is particularly important to consider in light of what we know about their distribution and concentration in plant material. Although the flavonoid composition of each plant species is generally so consistent that it has been used in taxonomic analyses, isoflavonoids differ in that they are known as phytoalexins, plant chemicals whose biosynthesis is dependent on the presence of plant stressors, especially plant pathogens. Consequently, isoflavonoid composition and concentration can be altered by a variety of environmental factors, including pathogens, soil nutrients and fertilizer, day length, cultivation under glass, temperature, and humidity (Whitten and Naftolin 1991). Therefore, it should be possible to both breed and cultivate plants to achieve the most beneficial mix of phytoestrogens, if we could determine what that mix might be. Moreover, because there are thousands of different flavonoids (Harborne 1979), sharing some but not all of the properties of isoflavonoids, it might be possible to identify a range of flavonoids that could be employed selectively in pharmaceutical use depending on the endpoint desired.

The findings presented here illustrate the broad range of actions of these natural estrogens and the variability in potency across endpoints. This variability argues for the importance of fully characterizing each phytoestrogen in terms of its sites of action, balance of agonistic and antagonistic properties, natural potency, and short-term and long-term effects.

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